A065 Triadic–Casimir / PKLSA Blind Falsifier v0.3.8 – Source-Native Core Preparation / Geometry-Derived Pilot

SST-Workbench; self-contained computational report

Run: A065-v0.4.3___20261020T011102486590Z-6f1d5bd0d3

Scientific status: This is a geometry and synthetic numerical pilot. Neither the time-dependent incompressible Euler equations nor a microscopic metal response nor a measured Casimir force has been certified. All preregistered failures are retained.

1 Identification and provenance

Run mode	FRAMEWORK
Framework	OFFICIAL_FRAMEWORK_EXECUTED
Blind campaign	SYNTHETIC_GATES_FAILED
PKLSA source points	300 KnotPlot trefoil (3_1)
Original source SHA-256	`a60b1a95a00110e3cb32087e...`

2 The three response channels

The June three-observable interpretation is retained as three distinct measurements, not as the sum of three forces:

$$\mathcal{O}[X;a] = \begin{pmatrix} \mathbf{a}_{\mathrm{bulk}}[X;a] \\ \Delta\nu/\nu[X;a] \\ P_{\mathrm{boundary}}[X;a] \end{pmatrix}.$$

With constant-density incompressible Euler kinematics,

$$\nabla\cdot\mathbf{u} = 0, \qquad \boldsymbol{\omega} = \nabla\times\mathbf{u}, \tag{1}$$

$$\nabla^2 p = \rho_M\left(\tfrac{1}{2}|\boldsymbol{\omega}|^2 - \|S\|_F^2\right), \qquad S_{ij} = \tfrac{1}{2}(\partial_i u_j + \partial_j u_i). \tag{2}$$

In a *stationary, irrotational* region the conditional Swirl-Clock reduction gives

$$p + \tfrac{1}{2}\rho_M|\mathbf{u}|^2 = C, \qquad N^2 = 1 - |\mathbf{u}|^2/c_\star^2 \implies \nabla\ln N = \frac{\nabla p}{\rho_M c_\star^2 N^2}.$$

This identity is conditional on selecting the same effective velocity in Bernoulli and the clock hypothesis; it is *not* a proof of gravitation.

3 Gauge-invariant clock difference and two-surface toy

The proposed coupling requires an as-yet-unverified gauge transformation of the atomic phases:

$$\Theta_{AB} = \theta_B - \theta_A - \sum_{e\in\mathcal{P}}\vartheta_e, \tag{3}$$

$$\dot\Theta_{AB} = \Omega_B^{(0)}N_B - \Omega_A^{(0)}N_A + \sum_{e\in\mathcal{P}}E_e. \tag{4}$$

In the *classical scalar test only* the two-sheet free-energy density is evaluated numerically from

$$\frac{\Delta\mathcal{F}}{A} = \frac{T}{4\pi}\int_0^\infty k\,\ln[1 - r_1 r_2\exp(-2ka)]\,dk.$$

The toy reflectivities r_j are model inputs and are not deduced from conducting metal atoms. There is no claimed quantum zero-point normalization.

4 PKLSA geometry and dynamical resolution

The source is a real $N = 300$ centerline, but the filament model is a reduced regularized ODE with RK4, not a 3D Euler PDE. All three spatial grids use the same original source, the same $\epsilon = 0.12$, $\Gamma = 1$, and $T = 0.04$. No grid-dependent core fitting is permitted. Its dimensionless velocity kernel is

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi} \sum_j \frac{\Delta \mathbf{X}_j \times (\mathbf{x} - \mathbf{X}_{j+\frac{1}{2}})}{\left(|\mathbf{x} - \mathbf{X}_{j+\frac{1}{2}}|^2 + \epsilon^2\right)^{3/2}}.$$

Quantity	Registered value
Final time T	0.04
Time-step relative discrepancy	2.0246251e-11
Spatial error (75 vs 150)	0.087289647
Spatial error (150 vs 300)	0.03266325
Spatial error (75 vs 300)	0.12122164
Predeclared spatial threshold	0.05
Historical 75–150 gate	**FAIL**

5 Independent source provider reconciliation

A single pre-run source manifest is verified by both the portable and official framework pipelines; each must recheck the exact raw bytes. A raw provider check is *geometry evidence only*, never an independent 3D Euler or Casimir-force certification.

Provider status	`PASS_RAW_HASH_AND_XYZ_PARSER_ONLY`
Provider SHA-256	`74cc9349324a15f4c1d4ca1a`
Provider points	118

6 Independently sealed datasets

The datasets use separate hash manifests. Different discretizations of the same original trefoil are *not* independent upstream providers.

Dataset	Seal digest (prefix)
D01_EULER	e79a4e0cfc370eb08a05d0288e8138a8...
D02_CLOCK_LINK	60729d7f5a206fdc025dbbde95d9284c...
D03_TWO_SURFACE	7d5e24b065ee9d6cbd43e3a9ce9a6d08...

7 Gate ledger

Gate	Status	Metric / reason
`D01_G01_EULER_DIV`	PASS	0.0
`D01_G02_POISSON`	PASS	0.0
`D01_G03_POTENTIAL_NULL`	PASS	0
`D01_G04_IRROTATIONAL_CLOCK_VARIATION`	PASS	0.0008849583697424501
`D01_G07_FILAMENT_RK4_DT_HALVING`	PASS	2.0246251170010892e-11
`D01_G08_FILAMENT_SPATIAL_REFINEMENT`	FAIL	0.08728964728469735
`D01_G09_FILAMENT_SPATIAL_150_300`	PASS	0.03266324959089548
`D01_G10_FILAMENT_SPATIAL_75_300`	FAIL	0.12122163649103088
`D01_G05_PKLSA_SOURCE_INTEGRITY`	PASS	300
`D01_G06_PKLSA_FINITE_BIOTSAVART`	PASS	0.002346617627425827
`D02_G01_GAUGE`	PASS	5.551115123125783e-17
`D02_G02_PHASE_RATE`	PASS	5.551115123125783e-17
`D02_G03_HOLONOMY`	PASS	2.0816681711721685e-17
`D02_G04_RESPONSE_POSITIVE`	PASS	0.0
`D02_G05_SHEET_STABLE`	PASS	1.6305756807341665
`D02_G06_NO_RESPONSE_CONTROL`	PASS	0.0
`D03_G01_ENERGY_STRESS`	PASS	1.141916836020315e-09
`D03_G02_DECOUPLED_NULL`	PASS	0.0
`D03_G03_FINITE_TOY_RESPONSE`	PASS	negative=28, total=28

Non-pass gates: D01_G08_FILAMENT_SPATIAL_REFINEMENT, D01_G10_FILAMENT_SPATIAL_75_300.

8 Two-surface synthetic fit

Pair	T	Fitted exponent
A/B	0.8	2.198691
A/B	1.6	2.198691
B/B	0.8	2.1234009
B/B	1.6	2.1234009

9 Qualification boundaries and reproducibility

- E010/Fremlin evidence is checked using one identical pre-run manifest in portable and framework mode. Verified raw geometry is not physical independent replication. No invented replacement is accepted.
- No calibrated moving metal surface, atom–link response, normal-mode quantum statistics, independently evaluated Casimir stress, or gravity-equivalence test is available.
- A065 physical discovery G3 is **UNRESOLVED / NOT RUN**; a successful program run cannot promote it to PASS.
- The frozen SST Falsifier Framework v1.0.7 remains external, unmodified, and authoritative when present. Its own TeX report and frozen gate ledger are kept in the per-run framework instance.
- This `.tex` contains **no** `\input`, `\include`, bibliography file, figures, or generated font dependencies. Recompile using `pdflatex` twice in the same folder.
- Each execution receives a UTC-stamped and random-suffixed run ID. Export files are opened create-only; previous experiments are never deleted or overwritten.

10 v0.3.2: Fixed-core spatial convergence and independent simulation provider

The historical 75/150 failure remains registered. The 600-node geometry is a periodic cubic interpolation of the same 300 PKLSA points, not independent source data.

$$X_{2j+1} = \frac{-X_{j-1} + 9X_j + 9X_{j+1} - X_{j+2}}{16}, \qquad X_{2j} = X_j.$$

For all spatial comparisons the original $T = 0.04$, $\Gamma = 1$, and $\epsilon = 0.12$ remain unchanged. The error statistic is the RMS difference of final node positions divided by the coarse-grid RMS displacement.

Pair	Relative error	Gate at 5%
75_vs_150	0.08728965	FAIL
150_vs_300	0.03266325	PASS
300_vs_600	0.008491986	PASS
75_vs_300	0.1212216	FAIL

Time refinement: $N = 300$, $4 \to 8$ steps: 1.691463e-09; $N = 600$, $8 \to 16$ steps: 8.115306e-11.

Independent Fremlin lane: SIMULATION_NUMERICAL_FAIL. Source label alone is not accepted: raw SHA-256, a separate C++17 executable, and spatial and temporal controls must all be verified. Missing raw geometry means unresolved, not a simulated pass.

Physical scope: these are two independent upstream geometries and independently implemented filament-ODE numerics only if Fremlin passes. They do not certify a second full Euler/PDE dynamics provider, Casimir stress, or gravitation.

11 v0.3.3: Gate-routing qualification

Historic G2 remains FAIL. The G3 execution decision uses only a precommitted G1/G10 refined numerical prerequisite and strict analytic/temporal/provenance guards; it is NOT a physical PASS.

Routing ready: True. G3 permitted outcome: UNRESOLVED. Physical force status: UNRESOLVED. Routing precommit SHA-256: 3856a587779d82df3b11ad1e64c4e4a45b6b4318e30c00676e4015d4df841fd6. GPU G6/G7 remain disabled in CPU profile.

12 v0.3.4 Historical trefoils: 2015 Fourier and torus seeds

Scope: Both samples are historical initial geometric curves, not relaxed PKLSA solutions, not a physical core or Euler validation. Their coordinate units are unscaled dimensionless values.

Precommit SHA-256: 73545eaddc62a3f2eda0553610220812758dc628ad2aa0eb8d4e9177d36207e5.

Numerical backend: EXECUTED.

12.1 A_Fourier_2015_p32

Historic page 32. Formula: [cos(t)+1.5*cos(2t), sin(t)-1.5*sin(2t), 0.35*sin(3t)].

Analytic length = 19.97652199; mean edge N600 = 0.0332934.

150_vs_300: 0.0332575 (PASS).

300_vs_600: 0.009163048 (PASS).

600_vs_1200: 0.002283298 (PASS).

Python/C++ parity (N150 RMS) = 4.544e-18.

$\epsilon = 0.06$: relative difference = 0.08872533.

$\epsilon = 0.24$: relative difference = 0.1566725.

12.2 B_Torus_2015_p66

Historic page 66. Formula: [(2+cos(3t))*cos(2t), (2+cos(3t))*sin(2t), sin(3t)].

Analytic length = 31.89860067; mean edge N600 = 0.0531623.

150_vs_300: 0.08112443 (FAIL).

300_vs_600: 0.02914099 (PASS).

600_vs_1200: 0.007459044 (PASS).

Python/C++ parity (N150 RMS) = 9.065e-18.

$\epsilon = 0.06$: relative difference = 0.06565456.

$\epsilon = 0.24$: relative difference = 0.1130557.

No physical force claims: gravity, Casimir, metal boundary conditions, mass, and full Euler PDE are UNRESOLVED.

13 v0.3.5 Direct analytical trefoils – high resolution

Numerical-only evidence: fixed regularized Biot–Savart midpoint filament ODE with classical RK4, not physical Euler PDE or Casimir/gravity closure.

Preregistration SHA-256: 2e36f33e328020fd0942d89573fda8a774f6e41e1e2a8245ef46edf33f23113a.

Four direct parametric arc-length samplings: 600, 1200, 2400, 4800 points.

Baseline $\varepsilon = 0.12$; diagnostic $\varepsilon \in \{0.06, 0.12, 0.24\}$. Threshold 1% baseline; 0.1% is a separately reported target, not a preregistered gate.

Native status: EXECUTED.

13.1 A_Fourier_2015_p32

Baseline spatial status: PASS; temporal: PASS; parity: PASS.

$\varepsilon = 0.06$; 1200_vs_2400: 0.211758% (PASS); 2400_vs_4800: 0.0528703% (PASS); 600_vs_1200: 0.849902% (PASS).

$\varepsilon = 0.12$; 1200_vs_2400: 0.0570087% (PASS); 2400_vs_4800: 0.0142476% (PASS); 600_vs_1200: 0.22833% (PASS).

$\varepsilon = 0.24$; 1200_vs_2400: 0.0160835% (PASS); 2400_vs_4800: 0.00402052% (PASS); 600_vs_1200: 0.0643564% (PASS).

13.2 B_Torus_2015_p66

Baseline spatial status: PASS; temporal: PASS; parity: PASS.

$\varepsilon = 0.06$; 1200_vs_2400: 0.679595% (PASS); 2400_vs_4800: 0.169058% (PASS); 600_vs_1200: 2.65588% (FAIL).

$\varepsilon = 0.12$; 1200_vs_2400: 0.185547% (PASS); 2400_vs_4800: 0.0463269% (PASS); 600_vs_1200: 0.745904% (PASS).

$\varepsilon = 0.24$; 1200_vs_2400: 0.0505952% (PASS); 2400_vs_4800: 0.0126449% (PASS); 600_vs_1200: 0.202633% (PASS).

Physical force, full Euler, true PKLSA dynamics: UNRESOLVED.